Jamhuriya University of Science and Technology - JUST
جامعة جمهورية للعلوم والتكنولوجيا

Final Project Guidelines

Advanced Java with Spring Boot

Project Type: Individual
Total Marks: 30
Project Proposal: 25/05/2025, Sunday
Deadline: 25/06/2025, Wednesday

Objective: The final project aims to demonstrate the students' understanding and application of the concepts learned throughout the course. Each student will design and implement a full-stack web application using Spring Boot for the backend and a suitable frontend technology.

Project Overview

Students are required to develop a web application that includes the following key components:

1. **Backend Development**: Utilize **Spring Boot** to create RESTful services.
2. **Frontend Development**: Design and implement a user-friendly interface using **React** framework
3. **Database Integration**: Use **PostgreSQL** for data persistence.
4. **Authentication and Authorization**: Implement user authentication and role-based access control.

Detailed Requirements

1. **Project Proposal**
   - Each student must submit a brief project proposal outlining their project idea, objectives, and main features. This proposal should be approved by me, ***Shuab Daud Ahmed***, before starting development.
2. **Backend Development**
   - **Spring Boot Setup**: Initialize a Spring Boot project with the necessary dependencies.

- **Entities and Relationships**: Define entities and relationships according to the application's requirements.
- **Repository Layer**: Create repositories to handle CRUD operations.
- **Service Layer**: Implement business logic in service classes.
- **Controller Layer**: Develop RESTful APIs to handle client requests.
- **Data Validation**: Ensure proper data validation using annotations.

3. **Frontend Development**
   - **Responsive Design**: The application must be responsive and accessible on different devices (desktops, tablets, and mobile phones).
   - **User Interface**: Create a clean, intuitive, and user-friendly interface.
   - **Forms and Validation**: Implement forms for data input with appropriate validation.
   - **Data Display**: Display data fetched from the backend using dynamic content rendering.
4. **Database Integration**
   - **Database Design**: Design the PostgreSQL database schema based on the application's requirements.
   - **Database Connection**: Configure the Spring Boot application to connect to the PostgreSQL database.
   - **Data Persistence**: Implement data persistence operations (CRUD) using Spring Data JPA.
5. **Authentication and Authorization**
   - **User Authentication**: Implement user registration and login functionality.
   - **Role-Based Access Control**: Define roles (e.g., admin, user) and restrict access to certain parts of the application based on user roles.

Project Deadline: 25/06/2024, Wednesday

Project Presentation: 27/06/2025, Friday